

comet (t_4). This is possible because the optimal flyby parameters (depth and inclination of the planetocentric hyperbola) can be computed directly from t_2 and t_4 , thus reducing the dimensionality of the problem.

3.1.1. Earth-GA segment

For a given flyby date, the ecliptic longitude of the GA planet is computed from the ephemeris as

$$l_2^F = \Omega^F + \omega^F + n^F(t_2 - t_p^F), \quad (1)$$

where n^F is the mean motion of the planet. The minimum eccentricity of the conics that connect Earth with the planet is determined by the condition of being able to reach the planet:

$$a_{\min} = \frac{r^E + r^F}{2} \rightarrow e_{\min} = 1 - \frac{r^E}{a_{\min}}. \quad (2)$$

The upper bound of eccentricity is set by the maximum characteristic energy ($C3$) that the launcher is capable of:

$$v_{1,\max} = \sqrt{C3} + v^E = \sqrt{C3} + \sqrt{\frac{\mu^S}{r^E}}, \quad (3)$$

$$E_{\max} = \frac{v_{1,\max}^2}{2} - \frac{\mu^S}{r^E} \rightarrow a_{\max} = \frac{-\mu^S}{2E_{\max}} \rightarrow e_{\max} = 1 - \frac{r^E}{a_{\max}}, \quad (4)$$

where E denotes the mechanical energy and μ is the gravitational parameter. Due to the condition of departure from the perihelion, the eccentricity determines the transfer angle between Earth and the planet

$$\cos(\theta_{12}) = \frac{1}{e} \left(\frac{p}{r^F} - 1 \right), \quad (5)$$

with $p = a(1 - e^2)$ denoting the semilatus rectum. Likewise, the transfer time t_{12} (only the elliptic case is shown for the sake of brevity) can be computed from e

$$\cos(E) = \frac{e + \cos(\theta_{12})}{1 + e \cos(\theta_{12})} \rightarrow M = E - e \sin(E) \rightarrow t_{12} = M \sqrt{\frac{a^3}{\mu^S}}, \quad (6)$$